

JULIAN DIAZ-GUTIERREZ

CONTACT INFORMATION

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EDUCATION

University of Illinois Urbana-Champaign Ph.D. in Agricultural and Applied Economics	2020-2026
Universidad EAFIT , Colombia Master of Science in Economics	2015-2016
Universidad Santo Tomás , Colombia B.A. in Economics (with honors)	2008-2012

FIELDS OF RESEARCH

Primary: Health Economics, Development Economics, Labor Economics.
Secondary: Environmental Economics.

RESEARCH EXPERIENCE

University of Illinois, Urbana-Champaign Research and Teaching Assistant, Department of Agricultural and Consumer Economics	2020-2025
The World Bank Research Analyst, Latin America and the Caribbean Chief Economist Office	2017-2020
The World Bank Consultant, South Asia Poverty Global Practice	2016-2017
Universidad EAFIT Research Assistant, Department of Economics	2015-2016
Centro de Estudios Regionales del Magdalena Medio (CER) Research Assistant	2013-2014

PUBLICATIONS

Canavire-Bacarreza, G., Diaz-Gutierrez, J. E., & Hanauer, M. M. (2018). Unintended consequences of conservation: Estimating the impact of protected areas on violence in Colombia. *Journal of Environmental Economics and Management*, 89, 46-70.

WORKING PAPERS

Automation, Economic Shocks, and Infant Mortality: Evidence from Mexico (Job Market Paper)

Abstract: In this paper, I investigate the impact of increased robotics in the United States on infant mortality rates in Mexico. Using a shift-share design that leverages variations in industrial robot usage and the employment composition of export-oriented *maquiladoras*, which predominantly employ

women, I find that regions with higher exposure to U.S. automation experienced a greater rise in infant mortality rates. The analysis shows that women in manufacturing faced larger job losses than men, leading to reduced household income and access to employer-provided healthcare. This economic strain forced many women into self-employment within manufacturing, reducing time for childcare. Additionally, I present evidence suggesting that automation may increase risky behaviors, such as drinking and smoking, among uninsured women of childbearing age. These findings highlight the complex relationship between technological advancements and public health outcomes, emphasizing the need for policymakers to consider the cross-border effects of automation on global health and employment.

Trade Effects on Substance Abuse: Evidence from Colombia's Liberalization *R&R at World Development*

Abstract: This paper examines the impact of trade liberalization on substance abuse. I study Colombia's 2010 unilateral tariff reform (Decree 4114), with immediate cuts concentrated in manufacturing. Using a Bartik-style measure of local tariff exposure and a continuous-treatment difference-in-differences design, I link the reform to municipality-level rates of substance-abuse hospitalizations and emergency-room visits over 2009–2014 from administrative health records. Municipalities more exposed to the tariff cut experienced larger increases in substance abuse. After correcting for multiple testing across the nine ICD-10 categories, multiple-drug use is the only category whose response remains individually robust, with a weaker effect on sedatives; the aggregate effect is present for adults aged 25–44 and is larger and more precisely estimated for males than for females. The evidence is consistent with a labor-market mechanism: the reform contracted formal employment in exposed tradable sectors, and the substance-abuse increase concentrates in the contributory health-insurance regime that covers formal employees. Drug-supply expansion and selective migration are not supported by the data as alternative mechanisms. The findings document downstream health costs of trade reform where social-protection infrastructure is limited.

Deportation and Fertility: Evidence from U.S. Immigration Enforcement in Mexico (with Mary Arends-Kuenning)

Abstract: We estimate the effect of U.S. deportations on fertility in Mexican origin municipalities. We exploit the staggered rollout of the Secure Communities program (2008–2014) across U.S. counties, combined with pre-existing migrant network linkages, to construct a shift-share measure of predicted deportation exposure at the municipality level. A one percentage point increase in the predicted deportee share raises the total fertility rate by 0.165, or 7 percent relative to the sample mean. The effect is concentrated among women aged 20–29 and among women in union. Decomposing by birth order, we find that the increase is driven entirely by higher-order births among partnered women, while first births and union formation rates are unaffected. These results point to a reunification mechanism: deportees—96 percent of whom are men of reproductive age—return to partners with whom they already have children, and the couple has an additional child. Female employment rises in response to the shock but earnings do not, consistent with an added worker effect that does not raise the opportunity cost of childbearing. The fertility and labor supply responses are complementary consequences of the same household shock.

POLICY WORK CONTRIBUTIONS

Going Viral: COVID-19 and the Accelerated Transformation of Jobs in Latin America and the Caribbean (2020). Latin America and Caribbean Studies. Washington, D.C. World Bank.

The Economy in the Time of Covid-19 (2020). Latin America and Caribbean Semiannual Report . Washington, DC. World Bank.

WORK IN PROGRESS

Long-Term Effects of Student Mobilizations on Academic Achievement and Labor Participation in Colombia (with Angelica Sanchez-Diaz).

Land Demining and Mental Health in Colombia.

SCHOLARSHIPS, GRANTS AND HONORS

ACDIS Summer Research Fellowship, UIUC (U.S. \$6,000) (2023)

Travel Grant, Department of ACE UIUC (U.S. \$500) (2022)

Latin American and Caribbean Enviromental Economics Program Research Grant (U.S. \$7,500) (2016)

Latin American and Caribbean Enviromental Economics Program travel grant (U.S. \$1,200) (2017)

Master's full scholarship at Universidad EAFIT (2015-2016)

Best second-year paper, Ph.D. program in Agricultural and Applied Economics, UIUC (2023)

Cum Laude Distinction, Universidad Santo Tomás (2012)

TEACHING EXPERIENCE

University of Illinois, Department of Agricultural and Consumer Economics

Teaching Assistant

- ACE 240: Personal Financial Planning (spring 2022, fall 2022, spring 2023, fall 2023, spring 2024)
- ACE 349: Risk Management and financial planning across the life cycle (spring 2023, spring 2024)
- ACE 449: Retirement and benefit planning (fall 2021, spring 2023, spring 2024)

CONFERENCES AND SEMINARS

2026-World Bank LAC Chief Economist Office and LACEA Labor Network Conference on Technological Change and Labor Markets in LAC. Washington, DC.

2025-Midwest International Economic Development Conference, UIUC; International Policy and Development (IPAD) Workshop, UIUC; WELAC (LACEA) Workshop on Gender and Household Economics, Lima, Peru.

2024- International Policy and Development (IPAD) Workshop, UIUC

2023- AAEA Annual Meeting. Washington, DC

2018-6th World Congress of Environmental and Resource Economics. Gothenburg Sweden

2017-Latin American and Caribbean Environmental Economics Association (LACEEP). Viña del Mar, Chile.

SKILLS AND LANGUAGES

Statistical Programming	Stata, R, Python
Software & Tools	ArcGIS, QGIS, LaTeX, Excel
Languages	Spanish (native), English (proficient)

PROFESSIONAL SERVICE

Referee for Economic Analysis Review, Department of Economics, School of Economics and Business at Universidad Alberto Hurtado.

Graduate student coordinator, UIUC International Policy and Development workshop (2023-2024)

REFERENCES

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